



Querying Data

Comparison Operators

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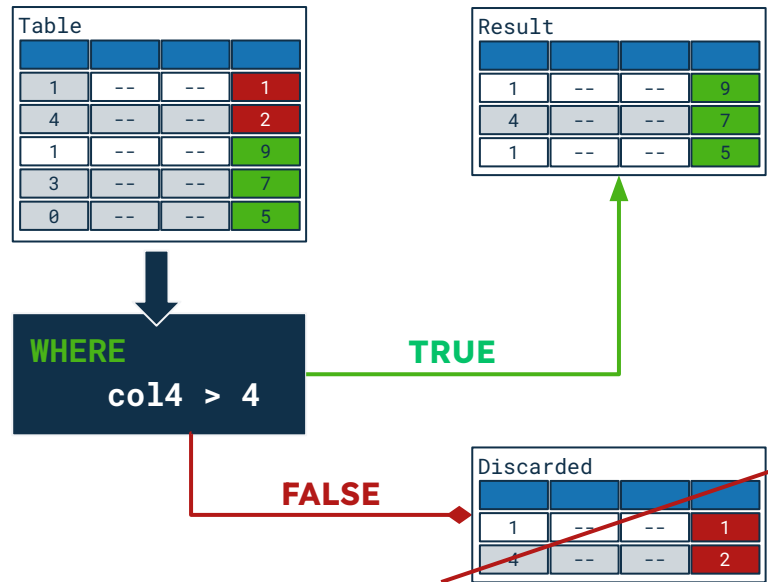
SQL Operators

Using the **WHERE** keyword, we can filter data. **Operators** can be used in conjunction with the **WHERE** clause to specify filter conditions.

Practical Uses:

- Find all countries with a gross domestic product (GDP) of **more than** \$400 billion.
- Find global regions with a population of **less than** 1 million.
- Find the development indexes **after** a certain date **and** for a **specific** country.
- Find elder death rates for sub-Saharan countries during the COVID-19 crisis (**between** 2020 and 2023).

For example, a **greater than** operator (**>**) in SQL only returns rows that have a value larger than 4 in col4.





Comparison Operators

Comparison operators are used to **compare values** and determine the truth or falsity of a condition based on the comparison result.

Examples of comparison operators in SQL include:

- equals (=)
- not equals (<>)
- less than (<)
- greater than (>)
- less than or equal to (<=)
- greater than or equal to (>=)

 =   A square is a triangle
→ **FALSE**

 ≠   A circle is not a triangle → **TRUE**

23 > 1  23 is greater than 1 → **TRUE**

"Kenya" = "Nigeria"  The strings "Kenya" and "Nigeria" are the same → **FALSE**



Comparison Operators

In SQL, comparison operators are commands used with **WHERE** that return the data in a row if the condition is **TRUE**.

=

Checks if **A = B**

Returns the rows where the value **A** in a column **equals** a specified value **B**.

>

Checks if **A > B**

Returns the rows where value **A** in a column is **greater than** a specified value **B**.

>=

Checks if **A ≥ B**

Returns the rows where value **A** in a column is **greater than or equal to** a specified value **B**.

<

Checks if **A < B**

Returns the rows where value **A** in a column is **less than** a specified value **B**.

<=

Checks if **A < B**
or
if **A ≤ B**

Returns the rows where value **A** in a column is **less than or equal to** a specified value **B**.

<>

Checks if **A ≠ B**

Returns the rows where value **A** in a column is **not equal** to a specified value **B**.



Our Example Database

We will use this placeholder database as an example:

Database (db)																																																								
<table><tr><th colspan="3">Table_1</th></tr><tr><th>col1</th><th>col2</th><th>col3</th></tr><tr><td>x</td><td>34</td><td>s</td></tr><tr><td>y</td><td>73</td><td>m</td></tr><tr><td>z</td><td>22</td><td>l</td></tr><tr><td>w</td><td>12</td><td>m</td></tr></table>	Table_1			col1	col2	col3	x	34	s	y	73	m	z	22	l	w	12	m	<table><tr><th colspan="3">Table_2</th></tr><tr><th>col1</th><th>col2</th><th>col3</th></tr><tr><td>car</td><td>68</td><td>s</td></tr><tr><td>cat</td><td>1</td><td>l</td></tr><tr><td>pet</td><td>7</td><td>l</td></tr><tr><td>cart</td><td>56</td><td>m</td></tr></table>	Table_2			col1	col2	col3	car	68	s	cat	1	l	pet	7	l	cart	56	m	<table><tr><th colspan="3">Table_3</th></tr><tr><th>col1</th><th>col2</th><th>col3</th></tr><tr><td>x</td><td>NULL</td><td>s</td></tr><tr><td>x</td><td>42</td><td>s</td></tr><tr><td>z</td><td>NULL</td><td>s</td></tr><tr><td>x</td><td>7</td><td>s</td></tr></table>	Table_3			col1	col2	col3	x	NULL	s	x	42	s	z	NULL	s	x	7	s
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Note: These are bad naming examples, but we use them here to show how queries work. Always try to use more descriptive titles.



Equal To

The = operator returns the rows where the value in a column **equals** a specified value.

```
SELECT
  *
FROM
  db.Table_2
WHERE
  col1 = "car";
```

Note that car is a **string**, so we enclose it in quotes.



col1	col2	col3
car	68	s
cat	1	1
pet	7	1
cart	56	m

col1	col2	col3
car	68	s
cat	1	1
pet	7	1
cart	56	m

TRUE

car is equal to car, so the row is **included**.

FALSE

cat, pet, and cart are not equal to car, so the rows are **excluded**.



Not Equal To

The operator `!=` or `<>` can be used to return the rows where the value in a column is **not equal to** a specified value. Both `!=` and `<>` serve as alternatives for expressing inequality.

```
SELECT
  *
FROM
  db.Table_2
WHERE
  col1 <> "car";
```

Note that we add a space before and after the operator in SQL code to make it more readable.



col1	col2	col3
car	68	s
cat	1	1
pet	7	1
cart	56	m

col1	col2	col3
car	68	s
cat	1	1
pet	7	1
cart	56	m

FALSE

car is equal to car, so the row is **excluded**.

TRUE

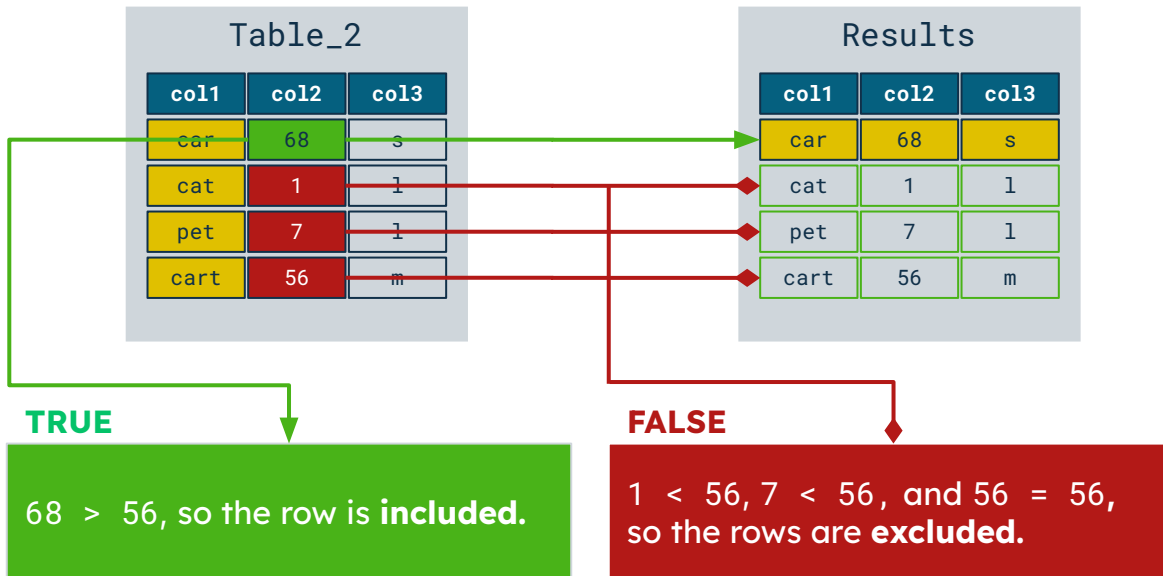
cat, pet, and cart are **not** equal to car, so the rows are **included**.



Greater Than

The `>` operator returns the rows where the value in a column is **greater than** a specified value.

```
SELECT
  *
FROM
  db.Table_2
WHERE
  col2 > 56 ;
```





Less Than or Equal To

The `<=` operator returns the rows where the value in a column is **less than or equal to** a specified value.

```
SELECT
  *
FROM
  db.Table_2
WHERE
  col2 <= 56;
```

Using `col2 <= 56` includes 56, unlike `<`.



col1	col2	col3
car	68	s
cat	1	l
pet	7	l
cart	56	m

col1	col2	col3
car	68	s
cat	1	l
pet	7	l
cart	56	m

FALSE

$68 \geq 56$, so the row is **excluded**.

TRUE

$1 \leq 56$, $7 \leq 56$, and $56 \leq 56$, so the rows are **included**.



Greater Than or Equal To

The `>=` operator returns the rows where the value in a column is **greater than or equal to** a specified value.

```
SELECT
  *
FROM
  db.Table_2
WHERE
  col2 >= 56;
```

